BUDA 535: Business Data Mining

Homework assignment 1

Instructor: Bin Liu

Due date: 03/22/2026 to eCampus

Rules

1. Submit your file in a word or pdf format to eCampus alone with your supporting files (eg., code, data) when necessary.
2. You can also prepare your answers in One Single Jupyter notebook file. You can include your answers in Markdown and Code cells.
3. When programming is required, please run all codes first and then submit the Jupyter notebook file that includes all outputs.
4. Write you answers clearly.

Problem 1 Decision trees [4 points]

Part I: Decision tree construction. As we introduced, entropy can be used to measure the impurity when splitting a node in a decision tree. Entropy is defined as

$$\text{Entropy} = -\sum_{i=0}^{C-1} p_i(t) \log_2 p_i(t),$$

where $p_i(t)$ is the frequency of class i at node t, and C is the total number of classes. Consider the training examples shown bellow for a binary classification problem.

Movie ID	Format	Movie Category	Class
1	DVD	Entertainment	C0
2	DVD	Comedy	C0
3	DVD	Documentaries	C0
4	DVD	Comedy	C0
5	DVD	Comedy	C0
6	DVD	Comedy	C0
7	Online	Comedy	C0
8	Online	Comedy	C0
9	Online	Comedy	C0
10	Online	Documentaries	C0
11	DVD	Comedy	C1
12	DVD	Entertainment	C1
13	Online	Entertainment	C1
14	Online	Documentaries	C1
15	Online	Documentaries	C1
16	Online	Documentaries	C1
17	Online	Documentaries	C1
18	Online	Entertainment	C1
19	Online	Documentaries	C1
20	Online	Documentaries	C1

Answer following questions:

a) Compute the Entropy for the **Movie ID** attribute.

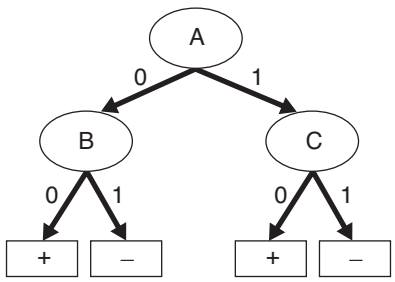
b) Compute the Entropy for the **Format** attribute.

c) Compute the Entropy for the **Movie Category** attribute using multiway split.

d) Which of the three attributes has the lowest Entropy?

e) Which of the three attributes will you use for splitting at the root node? Briefly explain your choice.

Part II: Decision tree evaluation. If you already built a decision tree as shown bellow ((a) on the left) for a binary classification problem. Then you apply the decision on a test data as shown in the table on the right.



(a) Decision tree

Instance	A	B	C	Class
1	0	0	0	+
2	0	1	1	+
3	1	1	0	+
4	1	0	1	-
5	1	0	0	+

(b) Test dataset

f) What's the accuracy of the decision tree model on the test data?

Problem 2 Linear models [4 points]
For logistic regression model, we fist have a decision function $z = b + w_1x_1 + w_2x_2 + ... + w_dx_d$; and we have an object assigned to positive example with a probability of $p(y = 1|\mathrm{x}) = \sigma(z) = \frac{1}{1+e^{-z}}$.

a) What are the **parameters** for a generic logistic regression model as follows?

$$z = b + w_1x_1 + w_2x_2 + ... + w_dx_d$$
$$p(y = 1|\mathrm{x}) = \sigma(z) = \frac{1}{1 + e^{-z}}.$$

b) Consider the following 15 points in a two-dimensional feature space (x_1, x_2) with labels as indicated (green $+$ is positive, red $\times$ is negative) in Figure 1. Let's assume that after training the logistic regression model, you get the decision function as follows:

$$z(x_1, x_2) = 4x_1 + 6x_2 - 24.$$

Answer the following questions:

b-1) What is the z value of the decision function for point A? And what's the probability of point A as a positive example?

b-2) In the figure above, draw the decision boundary, $z(x_1, x_2) = 0$, and indicate the positive and negative regions (half-planes).

b-3) What is the accuracy of this linear classifier?

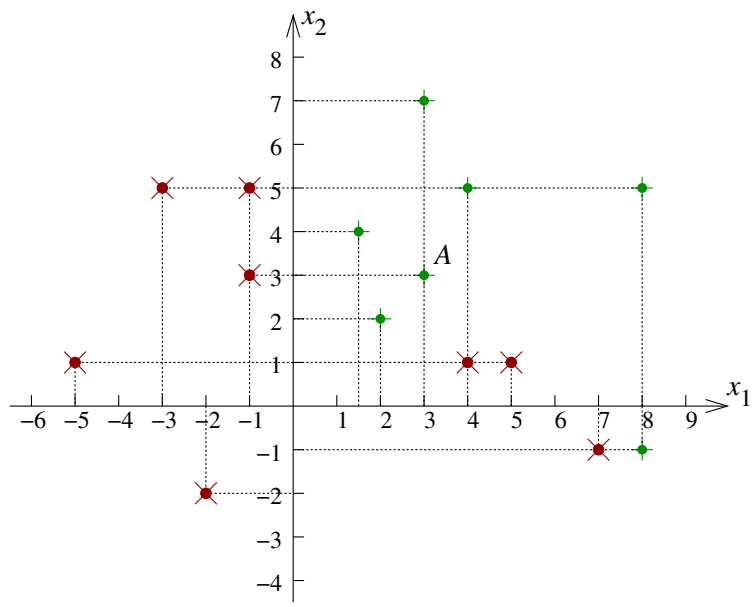


Figure 1: Data set for question 2

b-4) If you apply the model to a test dataset and get the confusion matrix as follows:

		Predicted class	
		+	-
Actual class	+	45	5
	-	10	40

i) Calculate the accuracy.
ii) Calculate the Precision and Recall.

Problem 3 Programming, Jupyter notebook template provided [4 points]
The goal of this question is to get you familiar with training and testing machine learning models with Scikit-learn library.

The Pima Indians Diabetes Dataset involves predicting the onset of diabetes within 5 years in Pima Indians given medical details. The dataset is originally from the National Institute of Diabetes and Digestive and Kidney Diseases. The objective of the dataset is to diagnostically predict whether or not a patient has diabetes, based on certain diagnostic measurements included in the dataset. It is a binary (2-class) classification problem. The number of observations for each class is not balanced. There are 768 observations with 8 input variables and 1 output variable. Missing values are believed to be encoded with zero values. Please refer the Jupyter notebook template for more details. In this question, you are given the *training data* and *test data*, which are derived from the 768 observations.

- Training data: `https://github.com/binbenliu/Teaching/blob/main/IntroAI/data/diabetes_train.csv`
- Test data: `https://github.com/binbenliu/Teaching/blob/main/IntroAI/data/diabetes_test.csv`

You are asked to train and test following models:

a) Decision trees
`https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClass`
`html#sklearn.tree.DecisionTreeClassifier`

b) Logistic regression
`https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.`
`LogisticRegression.html`

After you train each model on the *training data*, you should report following classification metrics on the *test data*: (1) accuracy, (2) precision, (3) recall, and (4) F1_score.

Also, (1) for Decision tree model, display the decision tree you derived, and discuss how decision tree make the classifications; and (2) for Logistic regression, display the feature importance scores, namely, $w_1, w_2, ..., w_d$, and discuss what you find from the model.

Your final submission should include the Python code in jupyter notebook to generate the classification results. Your grade will **NOT** be determined by the accuracy you are able to achieve, though we encourage you to explore different learning settings (e.g., choosing the right regularization strength) to get better results. Instead, we want to see you are able to train and test machine learning models with Scikit-learn library.